Age Sorting in the Labor Market

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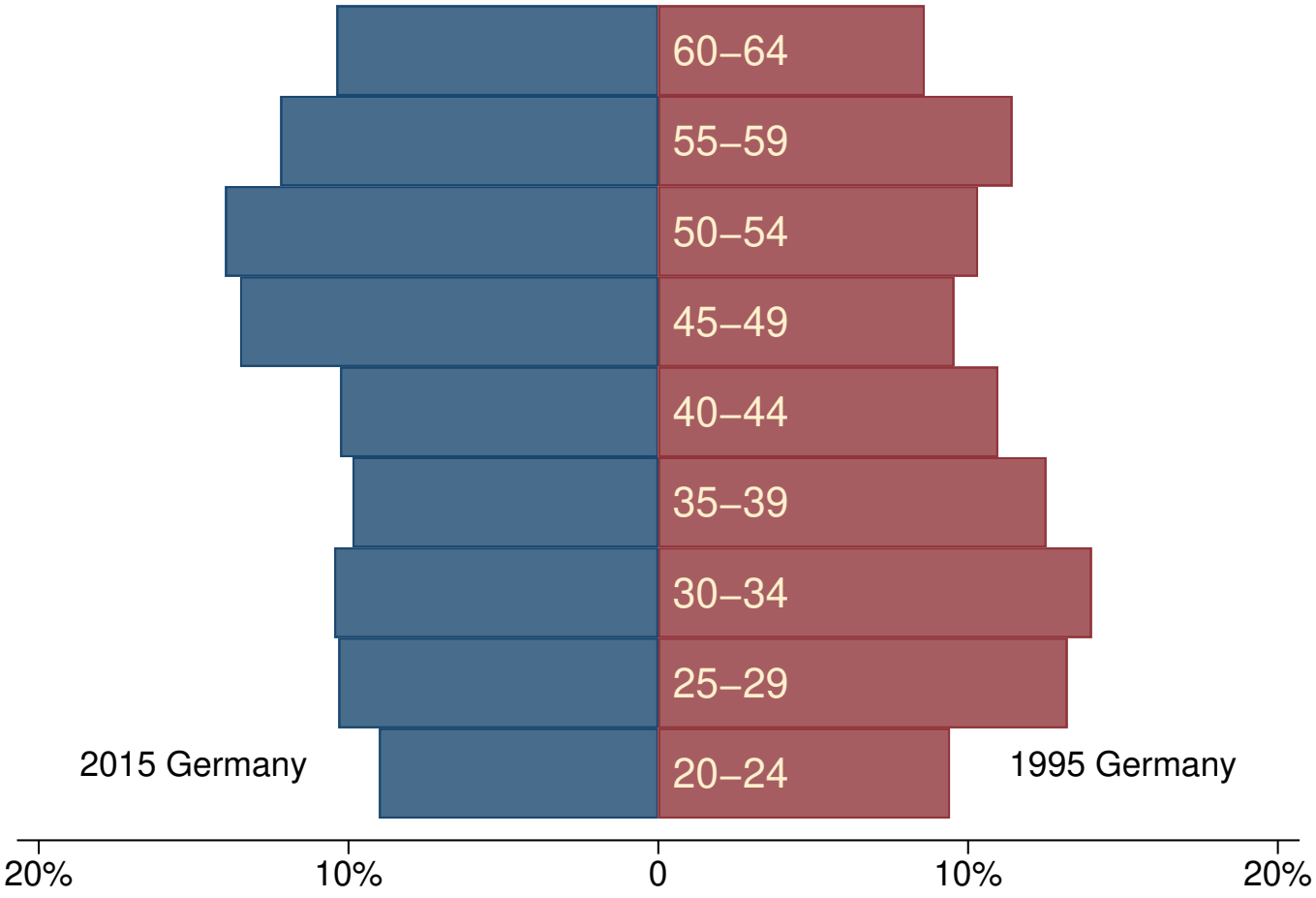
PRELIMINARY AND INCOMPLETE

Motivation: aging population



Motivation

- Aging of **labor force**:
  - declining fertility, share of young workers in the labor force
  - young workers are the main drivers of worker flows
- Aging of **firms**:
  - declining business dynamism, entry of new firms
  - young firms are the key players in creating new jobs
- How do these phenomena interact with each other?

  $\Longrightarrow$ what exactly are the roles of worker age and firm age in labor market?
  - just the proxies for heterogeneous productivity of workers/firms?
  - do younger workers disproportionally sort into younger firms?
  - young workers at young firms vs young workers at old firms?

This paper

- **Empirics:** evidence of age sorting in the labor market using German matched employer-employee data
  - younger firms hire disproportionately more younger workers
  - age sorting is not fully explained by sorting based on other observables (wage/industry/location/occupation etc.)

- **Theory:** model of worker life-cycle + firm life-cycle
  - equally-productive young firms and old firms are different in terms of the uncertainty/volatility about their future productivity
  - younger workers with longer life spans more likely to direct their search toward younger firms with higher volatility

- **Application:** quantify the impacts of aging labor force and firm on the labor market outcomes of current young population
  - change in age structure and decline in firm entry generate large decrease in employment rates among youth

Literature

- **Sorting in the labor market**:
  - Borovickova and Shimer (2020), Babina et al. (2019), Davis and Haltiwanger (2019), Bonhomme, Lamadon and Manresa (2019), Haltiwanger et al. (2018), Haltiwanger, Hyatt, and McEntarfer (2018), Lopes De Melo (2018), Ouimet and Zarutskie (2014), Card Heining and Kline (2013)
- **Worker age and transitions**:
  - Bayer and Kuhn (2020), Cajner, Guner, and Mukoyama (2020), Menzio et al. (2016), Gervais, Jamovich, Siu, Yedid-Levi (2016), Choi, Janiak, and Villena-Roldin (2015), Esteban-Pretel and Fujimoto (2014), Cheron, Hairault, and Langot (2013)
- **Firm dynamics and employment**:
  - Bilal, Engbom, Mongey, and Violante (2020), Schaal (2017), Kaas and Kircher (2015), Haltiwanger, Jarmin, and Miranda (2013), Elsby and Michaels (2013)

Data

Data

- Sample of Integrated Employer-Employee Data (SIEED)
  - 1.5% sample of all German establishments from 1975 to 2018
  - entire employment biographies of workers who have worked at those establishments
  - key variables: daily wage, employment spell, age, occupation, region + firm-level information
- Sample selection
  - non-financial private sector establishments with 1+ regular employee
  - workers aged 25-64 years old
  - focus on unemployment-to-employment transitions
- Define workers' **main jobs** for each quarter
  - establishment that offers the highest daily wage

Age Sorting in German Labor Market

Age sorting in German labor market

[Results pending disclosure review]

Model

Overview

- Model of worker life-cycle + firm life-cycle that generates age sorting
  - directed search model
  - workers heterogeneous in age
  - firms heterogeneous in productivity and underlying productivity uncertainty → less volatile as firms age

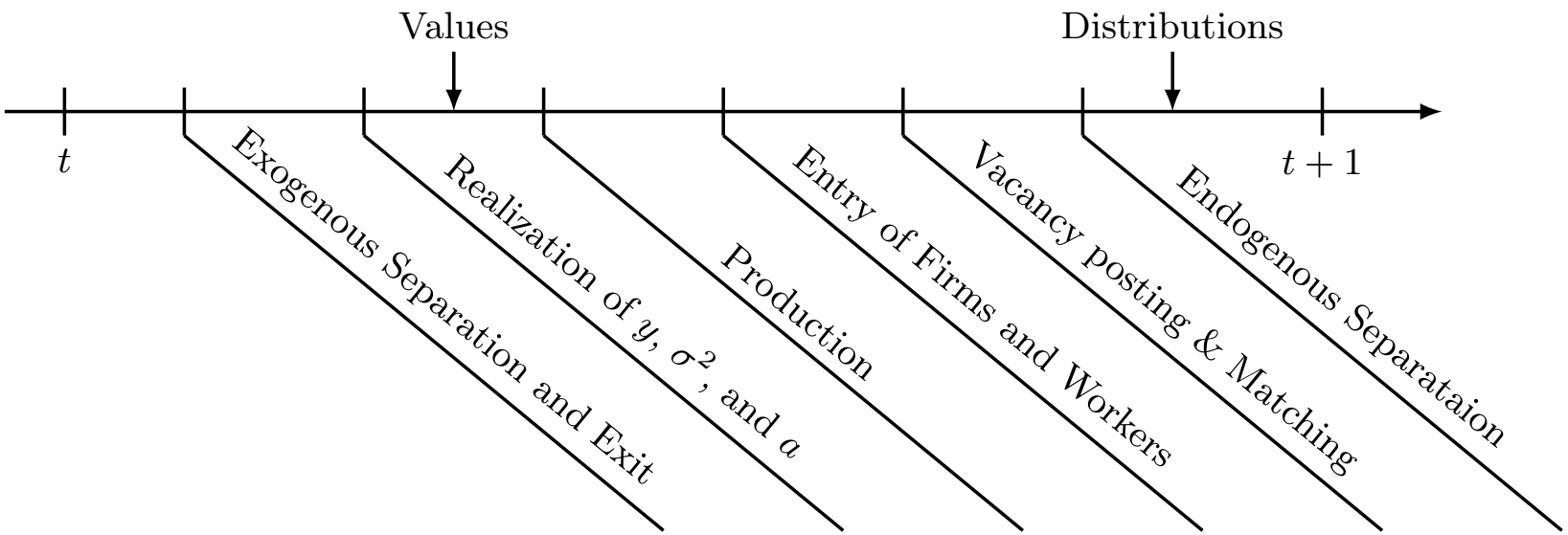
- **Key mechanism**
  - young firms are more uncertain about future productivity → conditional on current productivity, younger firms have more volatile productivity than older firms
  - young workers with longer life spans are more tolerant about the uncertainty and sort relatively more into younger firms

Model setup

- **Workers**
  - risk-neutral, employed or unemployed, discount future at rate R^{-1}
  - flow payoff b if unemployed
  - direct their search based on age and expected value
  - age with probability $\tau(a)$

- **Firms**
  - production linear in employment
  - idiosyncratic firm productivity following AR(1) process
  - volatility changes form σ_H to $\sigma_L (< \sigma_H)$ with probability p
  - post vacancies with convex vacancy costs
  - match can dissolve due to exogenous separation with probability δ_a, firm exit, or endogenous separation

Timing of decisions



Unemployed worker value

- Value of unemployed workers:

$$U(a) = b + \max_V \left\{ R^{-1} p(V,a)(V - \tau(a)U(a+1) - (1-\tau(a))U(a)) \right\}$$
$$+ R^{-1} \left\{ \tau(a)U(a+1) + (1-\tau(a))U(a) \right\}$$
$$U(A+1) = 0$$

- Workers are indifferent across different submarkets

$$\implies p(V,a) = \frac{U(a) - b - R^{-1}\tau(a)U(a+1) - R^{-1}(1-\tau(a))U(a)}{R^{-1}V - R^{-1}\tau(a)U(a+1) - R^{-1}(1-\tau(a))U(a)}$$

- Cobb-Douglass matching function $m(u,v) = Qu^{1-\eta}v^{\eta}$.

  $\implies$ vacancy-filling rate:

$$q(V,a) = Q^{\frac{1}{\eta}} p(V,a)^{\frac{\eta-1}{\eta}}$$

Firm value

- Firm value is decomposed into match and vacancy values.

$$\mathfrak{V}\left(y, \sigma^2, \{\omega_a\}_{a=1}^{A}\right) = \sum_a \omega_a W(y, \sigma^2, a) + X(y, \sigma^2)$$

- Value of a match:

$$\begin{aligned} W(y, \sigma_H^2, a) = y + R^{-1} \max\{ \int \Big[(1-p)\,(1-\delta_a)\Big(\tau(a) W(y', \sigma_H^2, a+1) \\ + (1-\tau(a)) W(y', \sigma_H^2, a)\Big) \\ + p\tau(a)(1-\delta_a)\Big(\tau(a) W(y', \sigma_L^2, a+1) \\ + (1-\tau(a)) W(y', \sigma_L^2, a)\Big)\Big] dF(y' | \rho \log y, \sigma_H^2) \\ + \delta_a \left(\tau(a) U(a+1) + (1-\tau(a)) U(a)\right), \\ R^{-1}\tau(a) U(a+1) + R^{-1}(1-\tau(a)) U(a)\} \end{aligned}$$

where $W(y, \sigma^2, A+1) = U(A+1) = 0$.

Expected match value

- Expected match value when posting a vacancy:

$$
\begin{aligned}
S(y,\sigma_H^2) := \int \Bigg[& (1-p)\Bigg\{\tau(a)\left(\tilde{W}(y',\sigma_H^2,a+1)-U(a+1)\right) \\
& + (1-\tau(a))\left(\tilde{W}(y',\sigma_H^2,a)-U(a)\right)\Bigg\} \\
& + p\Bigg\{\tau(a)\left(W(y',\sigma_L^2,a+1)-U(a+1)\right) \\
& + (1-\tau(a))\left(W(y,\sigma_L^2,a)-U(a)\right)\Bigg\}\Bigg] dF(y'|\rho\log y,\sigma_H^2)
\end{aligned}
$$

$$
\begin{aligned}
S(y,\sigma_L^2) := \int \Bigg\{ & \tau(a)\left(W(y',\sigma_L^2,a+1)-U(a+1)\right) \\
& + (1-\tau(a))\left(W(y',\sigma_L^2,a)-U(a)\right)\Bigg\} dF(y'|\rho\log y,\sigma_L^2)
\end{aligned}
$$

Vacancy value

- Value of vacancy posting for an active firm:

$$
\begin{aligned}
X\left(y, \sigma_H^2\right) = \max_{V(a) \geq U(a), m(a) \geq 0} & \Bigg\{ \sum_a m\left(a\right) q(V, a) R^{-1} \left(1 - \delta_a\right) \\
& \times \Bigg(\int \Bigg[(1-p) \left\{ \tau(a) \tilde{W}(y', \sigma_H^2, a+1) + (1 - \tau(a)) \tilde{W}(y', \sigma_H^2, a) \right\} \\
& + p \left\{ \tau(a) \tilde{W}(y', \sigma_L^2, a+1) + (1 - \tau(a)) \tilde{W}(y, \sigma_L^2, a) \right\} dF(y' | \rho \log y, \sigma_H^2) \Bigg] - V(a) \Bigg) \\
& + \sum_a \frac{m(a)}{\sum m(a')} \epsilon_a - c_v \left(\sum_a m(a) \right) \Bigg\}
\end{aligned}
$$

Directed vacancy posting over worker age

proof of theorem

Theorem 1

A firm with $(y, \sigma^2, \{\epsilon_a\})$ *chooses to post vacancies in only one submarket* (V^*, a^*).

$$V^* = (1-\eta)S(y, \sigma_L^2, a^*) + \tau(a)U(a^*+1) + (1-\tau(a))\,U(a^*)$$
$$a^* = \arg\max_a \left\{q(V,a)\,(1-\delta_a)\,R^{-1}S(y,\sigma^2,a) + \epsilon_a\right\},$$

Corollary 2

If $\epsilon_a \sim i.i.d.T1EV(\beta)$, *the probability that a firm targets to hire workers with* a *is given by*

$$\frac{exp\left(q(V,a)\,(1-\delta_a)\,R^{-1}S(y,\sigma^2,a)/\beta\right)}{\sum_{a'}\left(exp\left(q(V,a')\,(1-\delta_{a'})\,R^{-1}S(y,\sigma^2,a')\right)/\beta\right)}$$

Number of vacancies

- Number of vacancies posted by the firm with (y, σ^2) conditional on (V^*, a^*) satisfies

$$c_v'(m(y, \sigma^2, a^*)) = \eta R^{-1} \left(1 - \delta_{a^*}\right) q\left(V^*\right) S(y, \sigma^2, a^*)$$

  - by construction, this is independent of $\{\epsilon_a\}$

- Vacancy costs for entrants:

$$c_v^{e\prime}(m(y, \sigma^2, a^*)) = \eta R^{-1} \left(1 - \delta_{a^*}\right) q\left(V^*\right) S(y, \sigma^2, a^*)$$

  - Given the same states, entrants target the same age as incumbents, but the number of vacancies they post is higher.

Firm entry

- Free entry condition pins down the mass of entrants n_0 in the steady state.

$$f^e \geq \int X^e(y, \sigma_H^2) dH_0(y),$$

where $H_0(y)$ is the stationary distribution of the AR(1) process with σ_H^2, or $y \sim F(y|0, \sigma_H^2/(1-\rho^2))$.

Firm distribution

- Firm distribution over (y, σ^2):

$$
\begin{aligned}
H(y, \sigma_H^2) = 1 &\left\{ \max_a \left\{ S(y.\sigma_H^2, a) \right\} > 0 \right\} \\
&\times \left\{ (1-\chi)(1-p) \int_{y' \in R^+} F(y|\rho \log y', \sigma_H^2) dH(y', \sigma_H^2) \right. \\
&\left. + (1-\chi) n_0 H_0(y) \right\} \\
H(y, \sigma_L^2) = 1 &\left\{ \max_a \left\{ S(y.\sigma_L^2, a) \right\} > 0 \right\} \\
&\times \left\{ (1-\chi) \int_{y' \in R^+} F(y|\rho \log y', \sigma_L^2) dH(y', \sigma_L^2) \right. \\
&\left. + (1-\chi) p \int_{y' \in R^+} F(y|\rho \log y', \sigma_H^2) dH(y', \sigma_H^2) \right\},
\end{aligned}
$$

  where $\chi \leq \min_a \{\delta_a\}$ is exogenous firm probability.

- Firm exits the market if there is no worker with positive match surplus.

Worker distribution

- Worker distribution over (y, σ^2, a):

$$
\begin{aligned}
G(y, \sigma_H^2 | a) = {} & 1\left\{S(y.\sigma_H^2, a) > 0\right\}(1-\delta_a)(1-p)\,\tau(a-1) \\
& \times \int_{y' \in R^+} F(y|\rho \log y', \sigma_H^2) dG(y', \sigma_H^2 | a-1) \\
& + 1\left\{S(y, \sigma_H^2, a) > 0\right\}(1-\delta_a)(1-p)(1-\tau(a)) \\
& \times \int_{y' \in R^+} F(y|\rho \log y', \sigma_H^2) dG(y', \sigma_H^2 | a) \\
& + 1\left\{a(y, \sigma_H^2) = a\right\} m(y, \sigma_H^2) q(V(y, \sigma_H^2), a(y, \sigma_H^2))(H(y, \sigma_H^2) - n_0 H_0(y)) \\
& + 1\left\{a(y, \sigma_H^2) = a\right\} m^e(y, \sigma_H^2) q(V(y, \sigma_H^2), a(y, \sigma_H^2)) n_0 H_0(y)
\end{aligned}
$$

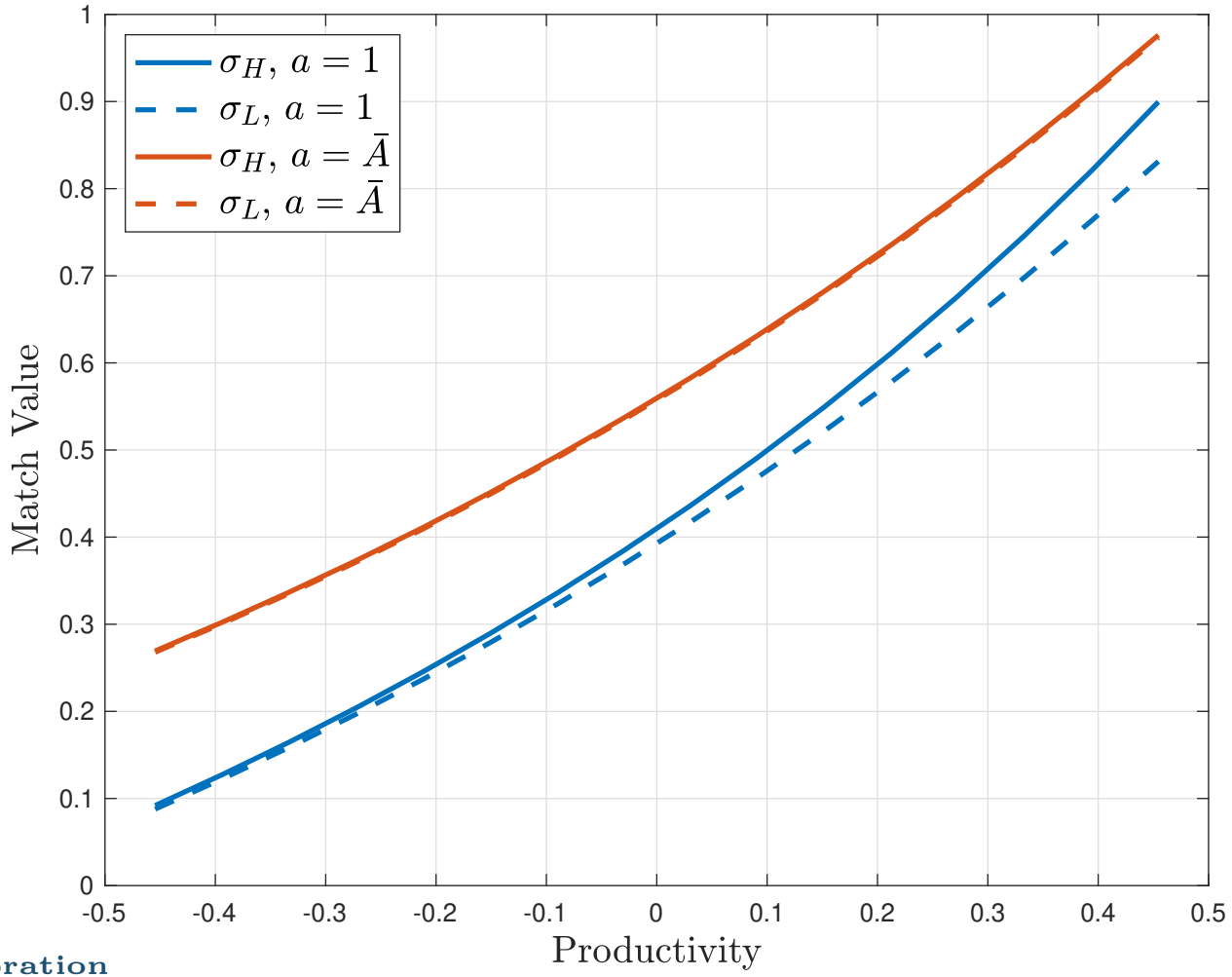
Equilibrium

- The stationary equilibrium is defined as firm and worker values $\{W(y, \sigma^2, a), U(a)\}$, firm policy function $V(y, \sigma^2), a(y, \sigma^2)$, distribution equation $H(y, \sigma^2, a)$ and $G(y, \sigma^2, a)$, and entry rate n_0 such that
  1. Given $\{U(a)\}$, firms optimize. Firms expect that the job-filling rates of unopen submarkets are consistent with the indifferent job-finding rates given $U(a)$.
  2. Distributions of firms and workers are time-invariant.
  3. Each submarket is cleared: i.e. the number of unemployed workers from the stationary distribution for g is same as the sum of the numbers of unemployed workers across submarkets at all ages.
  4. Free entry condition is satisfied.

Numerical Exploration

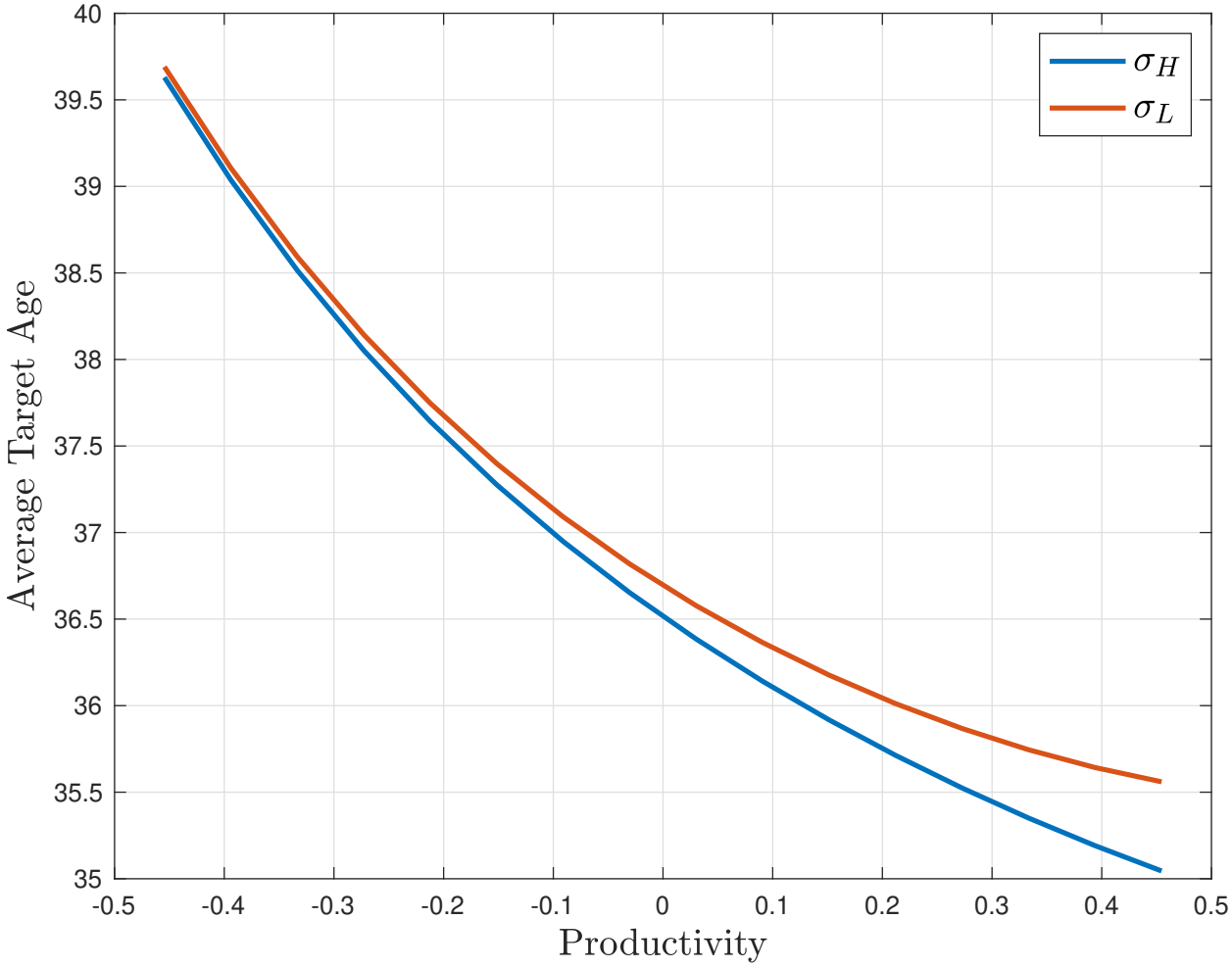
Value over age

- Younger workers have higher match values.
- A firm with higher σ^2 has a higher relative value of hiring younger workers.



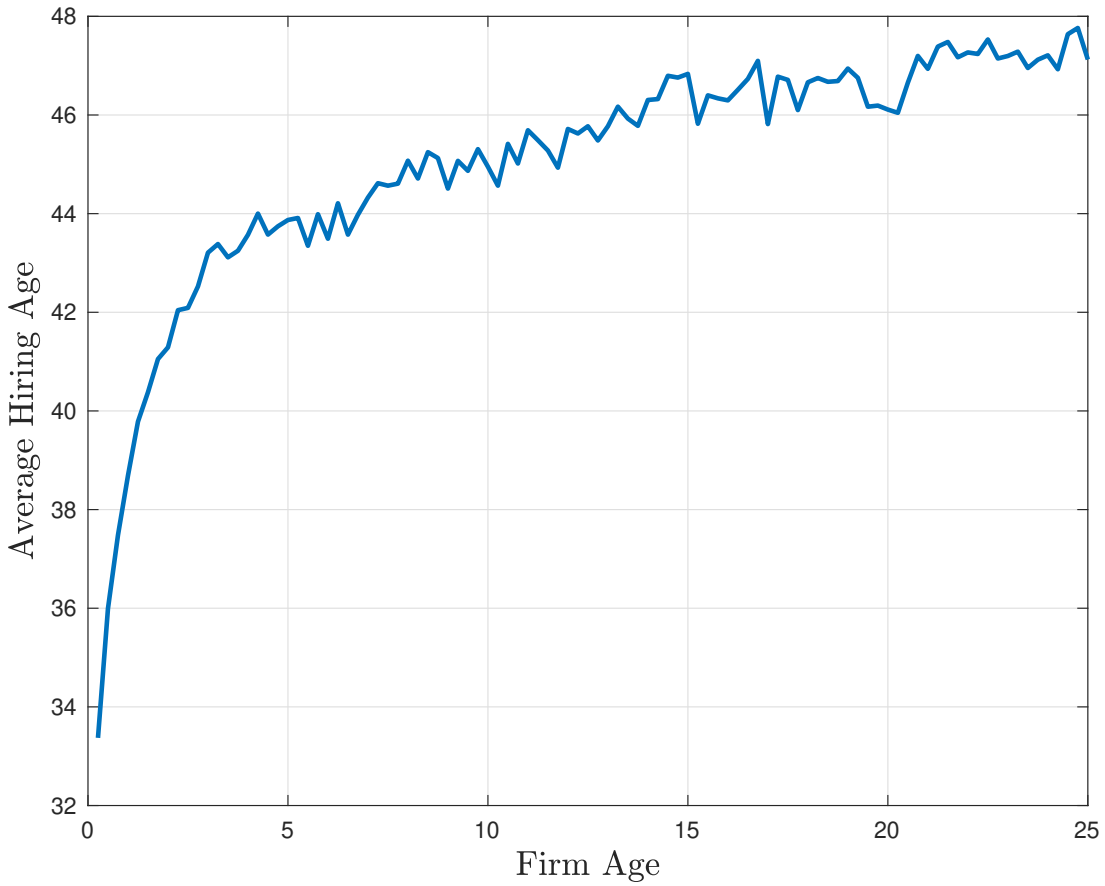
Targeted worker age

- A firm with higher volatility targets younger workers on average



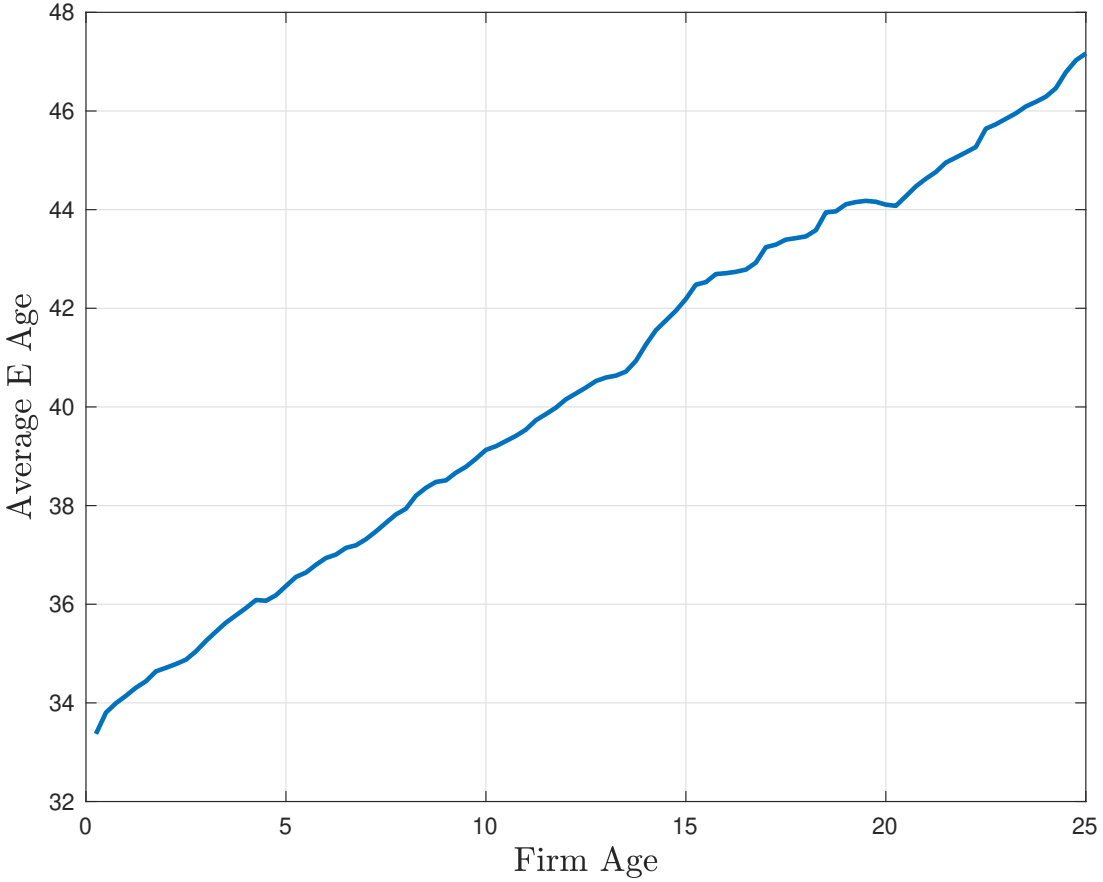
Firm age and worker age conditional on hiring

- Simulation delivers age sorting between firms and workers conditional on hiring.



Firm age and worker age conditional on employment

- Average worker age conditional on employment also features age sorting.



Calibration and Identification

Strategy

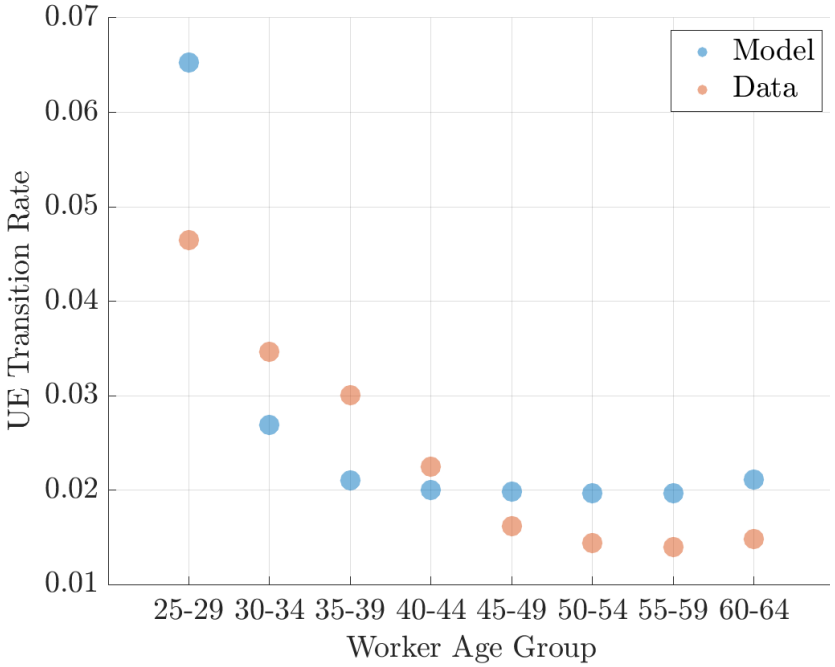
- Externally match the discount rate r and the population pyramid $\tau(a)$.
- Internally target aggregate, worker, & firm age-dependency moments in early 1995.
  - The aggregate unemployment rate. δ.
  - UE, EU rates for workers across worker age bins. η, Q, n_0, β, b.
  - Exit and growth rates for firms across est. age bins. p, σ_L, σ_H, ρ, γ, c_0, c_e.

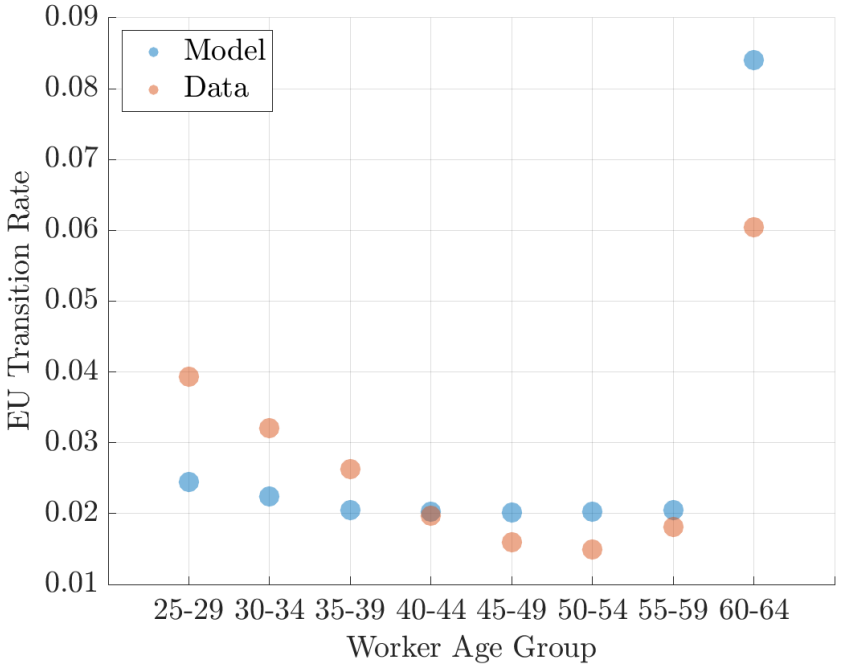
$\Rightarrow$ Validate the model by matching the age-sorting moments

- Average worker age conditional on hiring and empl. across est. age bins.

Results: targeted worker moments

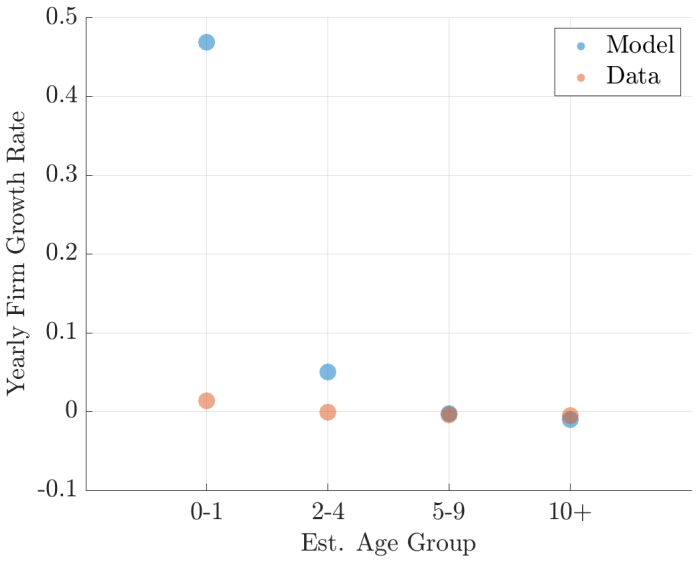
- Model matches the age dependency of EU/UE transition rates.

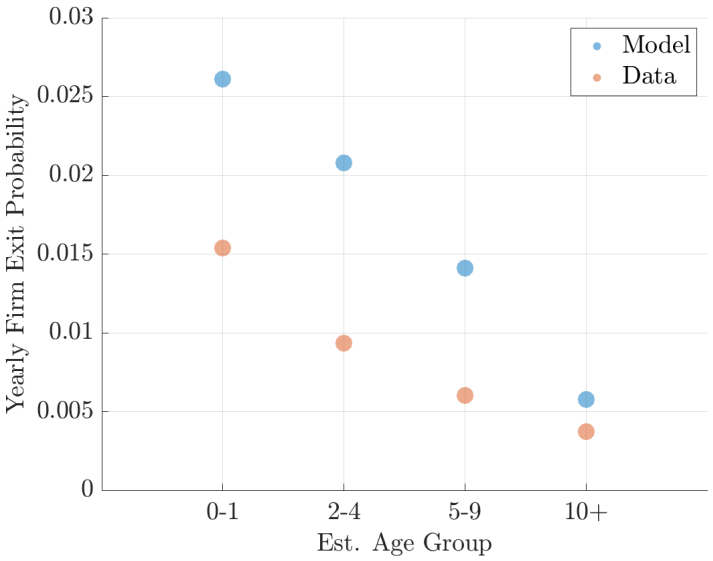




Results: targeted firm moments

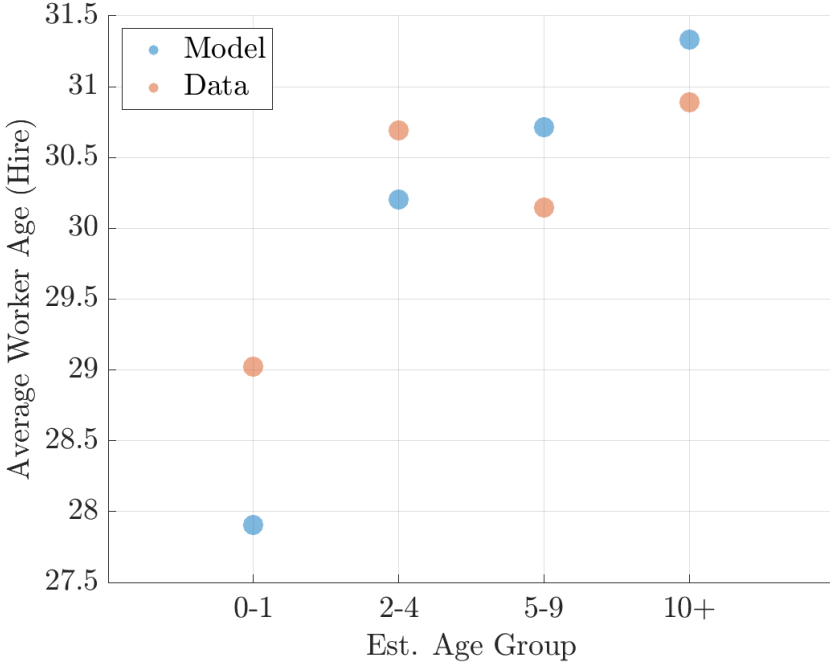
- Firm dynamics holds with a modest fit.

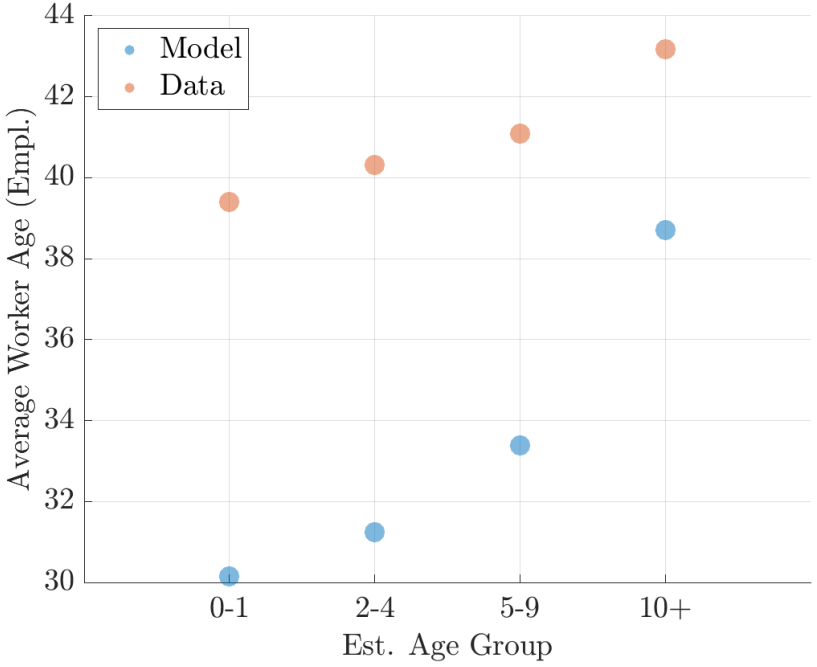




Results: untargeted age-sorting moments

- Calibrated model generates age sorting. Fit better for hiring than employment





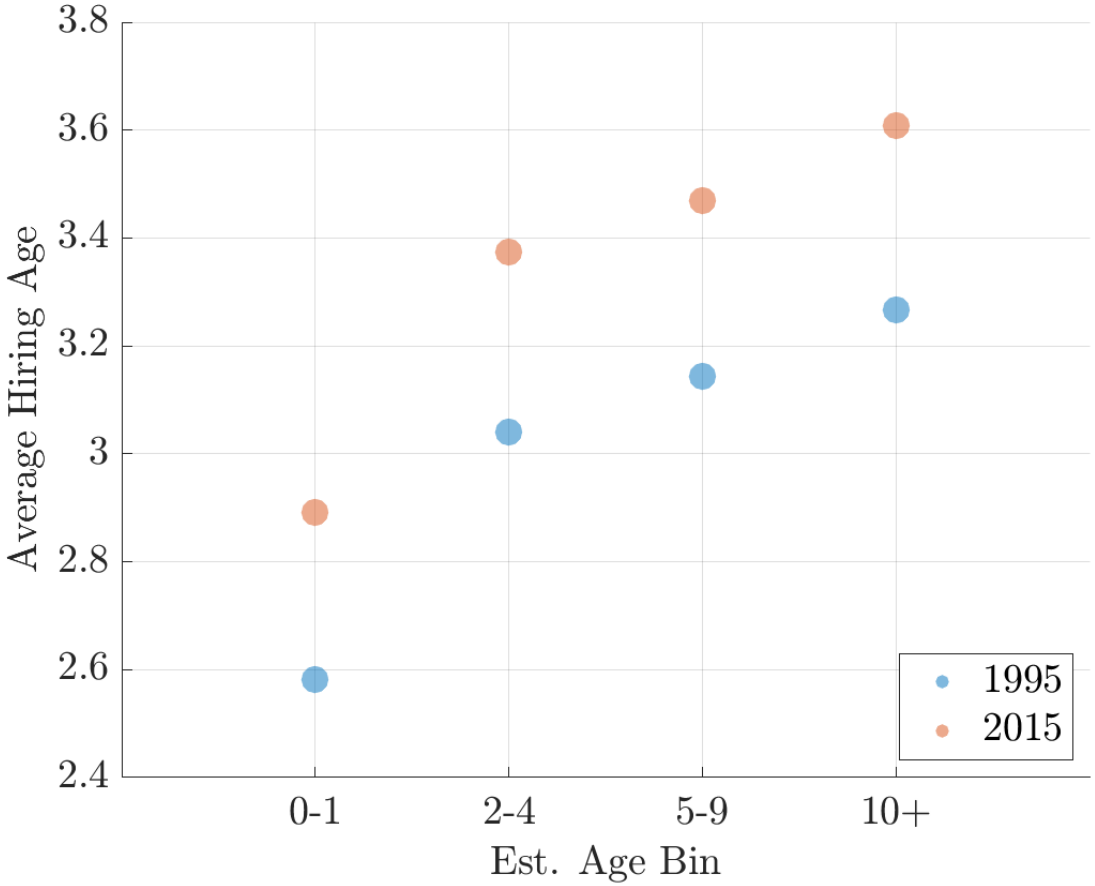
Counterfactual Analysis

Counterfactual analysis: youth non-employment

- Quantify the impacts of aging labor force and firm on the labor market outcomes of current young population
  1. start from the 1995 calibration parameters
  2. change the population structure ($\tau(a)$) as well as the entry rate (n_0) to 2015 values
  3. compute non-employment rates by worker ages

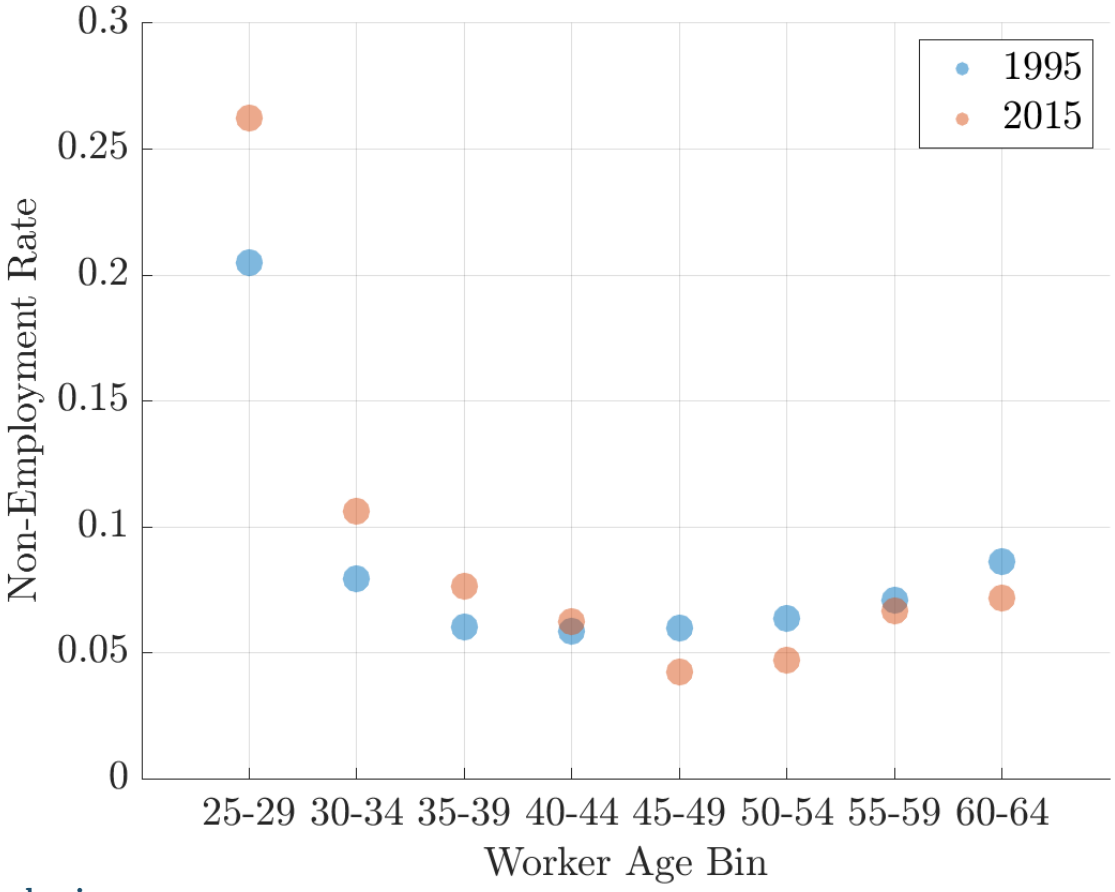
Counterfactual analysis: results

- Overall, average worker ages conditional on hiring are increasing



Counterfactual analysis: results

- Population aging and decline in entry rates increase non-employment rates for younger workers
- On the other hand, employment rates increase for older workers



Conclusion

- Young establishments are more likely to hire and work with young workers in Germany.
  - age sorting in UE, rather than EE, transitions plays a dominant role
  - young firm $\times$ young worker match has the lowest survival rate
- Model of worker and firm life-cycles that generates age sorting
  - young firms are more uncertain about future productivity.
  - young workers with longer life spans are more tolerant about the uncertainty
- Calibrated model matched the age-sorting patterns in UE transitions.
  - changes in firm entry rates and population structure generate a significant decrease in employment rates among young population

Appendix

Proof of Theorem 1 back

- The optimization problem is convex with convex costs and linear benefits.
- + The Kuhn-Tucker condition holds with the imposed solution.
- The first order condition with respect to $m(a)$ is

$$\left\{q(V,a)\left(1-\delta_{a^*}\right)R^{-1}\eta S(\mu,\sigma^2,a)\right\}-c_v'\left(\sum_a m(a)\right)+\frac{1}{\sum m(a')}\epsilon_a-\frac{\sum_a m(a)\epsilon_a}{\left(\sum m\left(a'\right)\right)^2}\leq 0$$

- Under the imposed solution, when $a \neq a^*$,

$$\eta q(V,a)\left(1-\delta_a\right)R^{-1}S(\mu,\sigma^2,a)-c_v'\left(m(a^*)\right)+\frac{\epsilon_a}{m(a^*)}-\frac{\epsilon_{a^*}}{m(a^*)}$$
$$<\eta q(V,a^*)\left(1-\delta_a\right)S(\mu,a^*)-c_v'\left(m(a^*)\right)=0.$$

$\square$